Summary of Revisions

Manuscript: A Comparability Protocol for Benchmarking Relational Database Access Layers in Express.js **Journal:** Information and Software Technology **Revision date:** 26 July 2026

Semantic admission

- Added an expected-result oracle derived from the deterministic seed and fan-out specification, independent of all adapters and both DBMSs.
- Kept differential comparison as a separate equivalence layer and renamed the protocol stage accordingly.
- Added exact campaign-state and allocator preflight checks.
- Added a 662-row cross-engine seed-parity gate; it caught differing fan-out comment-author assignments hidden by equal counts, and the affected pilot was rejected before the campaign restarted from repetition 1.
- Corrected a MySQL reset defect discovered by the new oracle and remeasured all affected patterns.

Treatment and implementation validity

- Renamed the case-study estimand "policy-selected documented configuration" and removed the behavioral developer persona.
- Made the selection policy substitutable at the protocol level.
- Preserved exact documentation responses, timestamps, hashes, conflict resolution, and alternatives.
- Added implementation-review provenance as recommended stage R7 and result-blind review packets.
- Marked independent human selection and adapter review as pending rather than claiming agreement.
- Recorded a result-blind author self-audit covering every adapter. It caught and removed per-insert Knex/MySQL warning I/O and an unused per-request MikroORM aggregation accessor; every affected pilot was preserved and rejected, the full admission chain was rerun, and measurement restarted from repetition 1. This is not counted as independent R7 evidence.

RQ2 and new measurements

- Ran one coherent corrected-state primary campaign covering all 18 compatible treatment–backend pairs for all five access patterns: 90 complete cells x 25 runs, zero timed request failures, and pre/post state checks.
- Ran a second independently ordered same-host campaign on 7 portable layers x 2 stacks x 3 prioritized patterns: 42 complete cells x 25 runs, zero timed failures. Deep-fetch ranks reproduce exactly; MySQL insert reproduces only 3/7 ranks and is reported as unstable.
- Predeclared persistence criteria retain four material cross-stack reversals and classify all close/nonrecurring changes as exploratory.
- Defined RQ2 as backend-stack transfer within the tested same-host campaigns.
- Removed causal DBMS attribution and made close rank reversals exploratory.
- Retained the explicit limitation that the second campaign is same-host, not cross-host replication.

Protocol contribution

- Narrowed novelty to a concrete domain-specific operational discipline built from established controls.
- Cited and distinguished CYNTHIA and retained the direct prior-art boundary.
- Added the 2023 Node-Bench comparison as a direct coverage precedent and a ninth source in the protocol audit.
- Added a seven-stage, nine-source retrospective protocol audit with 63 source-located judgments and a machine validator.
- Described that audit as a one-rater bounded applicability demonstration, not independent validation.
- Removed universal completeness and necessity language; each level now states the interpretation it licenses.

Fresh secondary analyses

- Reran common-SQL (16 cells x 10), all-pattern capacity (720 points), matched-utilization (64 cells x 5), longer-tail (18 cells x 10 x 60 s), relaxed durability (18 cells x 5), wait-event, and canonical-constructor analyses with zero accepted request failures.
- Updated all numerical claims: selected deep-fetch spread 7.01x/5.12x; common-SQL 1.71x/2.03x; stable 50% p99 bands 2.3–4.9/1.9–5.5 ms.
- Reframed concurrent wait sums and durability changes as compound sensitivity evidence, not a DBMS floor or causal share.

Reproducibility

- Reconciled the main paper with Supplement S31.
- Completed an isolated current-candidate reconstruction: 60/60 JSON hashes verified, and 35 table outputs plus four derived analysis JSON files regenerated byte-for-byte without starting either DBMS.
- Preserved and verified exact 42-file source archives for each accepted campaign, so later harness maintenance cannot overwrite the code provenance of completed measurements.
- Fixed the standalone write-admission cleanup to restore post and comment allocators even after failure; the corrected verifier passes all 18 adapter–engine admissions and leaves both campaign states exact.
- Named the five presentation-only differences in the historical 45/50 reconstruction.
- Separated author-run offline reconstruction, historical archive-isolated reconstruction, current measurement re-execution, and independent reproduction.
- Marked the Docker path as workflow reproduction because its durability and topology differ from the reference numerical path.
- Updated the manifest so every revised output maps to its exact data and generator.

Presentation

- Replaced “strategy standardization” with “common-SQL raw-path sensitivity contrast.”
- Restricted matched-utilization conclusions to stable 50% and most 70% cells.
- Regenerated the MySQL insert distribution from corrected state.
- Standardized “deep fetch” and “single-row insert.”
- Consolidated repeated qualifications and moved exploratory interpretation out of the main argument.